

DPABI: Data Processing & Analysis for (Resting-State) Brain Imaging

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Abstract Brain imaging efforts are being increasingly devoted to decode the functioning of the human brain. Among neuroimaging techniques, resting-state fMRI (R-fMRI) is currently expanding exponentially. Beyond the general neuroimaging analysis packages (e.g., SPM, AFNI and FSL), REST and DPARSF were developed to meet the increasing need of user-friendly toolboxes for R-fMRI data processing. To address recently identified methodological challenges of R-fMRI, we introduce the newly developed toolbox, DPABI, which was evolved from REST and DPARSF. DPABI incorporates recent research advances on head motion control and measurement standardization, thus allowing users to evaluate results using stringent control strategies. DPABI also emphasizes test-retest reliability and quality control of data processing. Furthermore, DPABI provides a user-friendly pipeline

analysis toolkit for rat/monkey R-fMRI data analysis to reflect the rapid advances in animal imaging. In addition, DPABI includes preprocessing modules for task-based fMRI, voxel-based morphometry analysis, statistical analysis and results viewing. DPABI is designed to make data analysis require fewer manual operations, be less time-consuming, have a lower skill requirement, a smaller risk of inadvertent mistakes, and be more comparable across studies. We anticipate this open-source toolbox will assist novices and expert users alike and continue to support advancing R-fMRI methodology and its application to clinical translational studies.

Keywords Data processing · Quality control · Resting-state fMRI · Standardization · Statistical analysis

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Introduction

Human brain function, as implemented by one of the most complex systems in the universe, is fascinating to decode. Brain imaging has advanced our progress in such efforts during recent decades. Among neuroimaging techniques, magnetic resonance imaging (MRI), especially functional MRI (fMRI), dominates because of its non-invasiveness, whole-brain coverage and wide availability (Bandettini 2012). By detecting local concentrations of deoxygenated and oxygenated hemoglobin, fMRI blood-oxygen level dependent (BOLD) signals allow inferring blood flow changes associated with differential neural activity during distinct tasks (Huettel et al. 2004). Unlike task-based fMRI requiring specific tasks tailored to each cognitive experiment design, resting-state fMRI (R-fMRI) focuses on spontaneous brain activity without requiring subjects to perform explicit cognitive tasks, other than remaining extremely still or ‘resting’ in the scanner (Biswal et al. 1995). R-fMRI has become increasingly popular

(Buckner et al. 2013; Fox and Raichle 2007; Kelly et al. 2012) because of its substantial reliability (Shehzad et al. 2009; Zuo and Xing 2014), sensitivity (Castellanos et al. 2013; Fox and Greicius 2010; Greicius 2008) and comparability across labs (ADHD-200-Consortium 2012; Biswal et al. 2010; Di Martino et al. 2014; Mennes et al. 2013). The R-fMRI field has been expanding exponentially, with more than 1,000 studies now being published per year (Bim 2012; Yan et al. 2014; Zuo and Xing 2014).

As the R-fMRI field has grown, there is an increasing need for user-friendly toolboxes for R-fMRI data processing that go beyond manually integrating functions in general analysis packages such as Statistical Parametric Mapping (SPM) (Ashburner 2012), FMRIB Software Library (FSL) (Jenkinson et al. 2012), or Analysis of Functional NeuroImages (AFNI) (Cox 2012). Song and colleagues developed a toolbox named Resting-State fMRI Data Analysis Toolkit (REST) (Song et al. 2011) to calculate common R-fMRI derivatives. Based on SPM and REST functions, a MATLAB toolbox for “pipeline” data analysis of R-fMRI data, DPARSF was also developed (Yan and Zang 2010). The DPARSF analysis pipeline automated and overcame time-consuming manual procedures in preprocessing R-fMRI data and generating R-fMRI derivatives, e.g., functional connectivity (Biswal et al. 1995), regional homogeneity (ReHo) (Zang et al. 2004), amplitude of low frequency fluctuations (ALFF) (Zang et al. 2007), fractional ALFF (fALFF) (Zou et al. 2008), voxel-mirrored homotopic connectivity (VMHC) (Zuo et al. 2010b) and degree centrality (Buckner et al. 2009). This user-friendly toolbox made R-fMRI analysis easily accessible to a broader community, as attested by its having been cited more than 600 times per Google Scholar. Although DPARSF has been widely used (including more recent R-fMRI pipelines CONN (Whitfield-Gabrieli and Nieto-Castanon 2012) and FATCAT (Taylor and Saad 2013)), the original DPARSF has fallen behind with some recent developments of R-fMRI methodologies, particularly in relation to R-fMRI methodological challenges.

For example, head motion is a formidable challenge for R-fMRI, as recent work conclusively demonstrated that even head micro-movements could introduce artifactual inter-individual differences in R-fMRI metrics (Power et al. 2012a, b; Satterthwaite et al. 2012; Van Dijk et al. 2012). Several studies (Fair et al. 2012; Power et al. 2014; Satterthwaite et al. 2013; Yan et al. 2013a) provided comprehensive assessments of motion-related artifacts and suggested procedures for controlling them, including higher-order regression models (e.g., Friston 24-parameter model (Friston et al. 1996)) and motion scrubbing approaches at the individual-subject level (Power et al. 2012a), as well as accounting for head motion at the group-level (i.e., covariate analysis). Such strategies need to be seamlessly integrated into

an easy-to-use R-fMRI pipeline toolbox. See a comprehensive review on recent advances regarding this topic in Power et al. (2015).

Measurement standardization represents another key challenge for R-fMRI. Concerns about the many sources of nuisance variation impacting on R-fMRI measures continue to grow (Cole et al. 2010; Kelly et al. 2012). A recent study compared an array of standardization approaches and found group-level post-hoc standardizations for mean-regression and variance-standardization were advantageous for avoiding the introduction of artifactual relationships with standardization parameters (Yan et al. 2013b). To better generalize such standardization to minimize the influence of nuisance variables in future studies, a user-friendly module for standardizing R-fMRI indices is warranted. Furthermore, to yield valid results, R-fMRI measures must have adequate test-retest (TRT) reliability (Shehzad et al. 2009; Zuo et al. 2010a; Zuo and Xing 2014). A module for assessing TRT reliability is also needed to facilitate investigating effects of inter-individual differences on R-fMRI measures.

With all the methodological improvements being incorporated into R-fMRI, adequate quality control (QC) is essential for establishing reliable results. For most advanced MRI centers, regular quality assurance protocols ensuring scanner performance stability are performed daily (Friedman and Glover 2006b; Ihalaenen et al. 2004; Simmons et al. 1999). Even when the quality of fMRI acquisition is monitored, quality control of data processing needs to be considered. The quality of raw structural and functional images, as well as the effects of registration to standard templates must be checked. Furthermore, spatial coverage of functional images and head motion during scanning should be considered in the quality control steps. Manual data checking is extremely time consuming, which sometimes prevents investigators from checking each subject's data. Thus, an integrated and efficient quality control module for data processing would be essential for ensuring the reliability of R-fMRI study findings.

Most R-fMRI studies are performed on humans. However, as the physiology underlying fMRI effects remains largely unknown, R-fMRI studies with animal models are critical (Hutchison and Everling 2012). Animal work, including rodents and non-human primates, allows neurophysiological recordings and pharmacological/behavior manipulations, which may not be ethical or practical in humans. There exist structural image analysis pipelines for animals; for example, SPMouse facilitates voxel-based morphometry analysis for mouse data (Sawiak et al. 2009). However, there is no user-friendly R-fMRI analysis pipeline designed for animal data to date.

To meet these demands, we developed a toolbox named DPABI (for Data Processing & Analysis of Brain Imaging) (Fig. 1), which evolved from DPARSF and REST. DPABI includes an advanced edition of DPARSF, which integrates

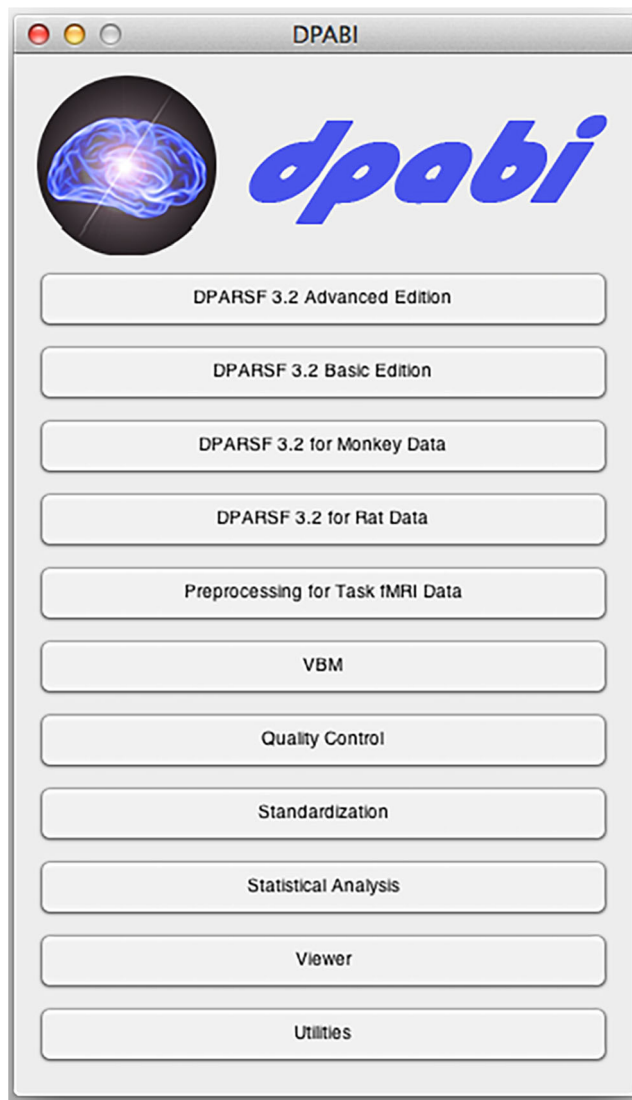


Fig. 1 DPABI's graphical user interface

the latest algorithms to control for head motion, as well as standardization, TRT reliability, quality control modules and rat/monkey data analysis pipelines. DPABI works with other imaging modalities, including preprocessing modules for task-based fMRI and for voxel-based morphometry analysis, as well as modules for statistical analysis. DPABI also includes a MRI viewer.

Implementation

DPABI (RRID:SCR_010501) was developed in MATLAB (The MathWorks Inc., Natick, MA, US). It is an open-source package (source code available at <http://rfmri.org/DPABI>) evolved from DPARSF and REST. It supports Windows, MacOS and Linux operating systems. Here, we introduce the implementation of its key features.

Advanced DPARSF

The implementation of DPARSF V1.0 (basic edition) was previously described (Yan and Zang 2010). The updated edition of DPARSF has been integrated into DPABI as a key module. Here we present the implementation of the novel functions in the recent edition (V3.2) as follows.

In addressing head motion concerns in R-fMRI analyses (Power et al. 2012a, b; Satterthwaite et al. 2012; Van Dijk et al. 2012), DPARSF V3.2 provides Friston 24-parameter motion correction (Satterthwaite et al. 2013; Yan et al. 2013a), i.e., regression with autoregressive models of motion incorporating 6 head motion parameters, 6 head motion parameters one time point before, and the 12 corresponding squared items (Friston et al. 1996). DPARSF V3.2 also provides voxel-specific head motion calculation and correction at the individual level (Satterthwaite et al. 2013; Yan et al. 2013a). Data scrubbing was implemented in multiple ways: 1) model each bad time point as a separate regressor in nuisance covariates regression, 2) delete bad time points, and 3) interpolate bad time points with nearest neighbor, linear or cubic spline interpolation. The bad time points can be detected according to their framewise displacement (FD) value, either by the measure defined by Power et al. (2012a) or the one defined by Jenkinson et al. (2002). The bad time points could be flagged by any volume with $FD > 0.2$ mm (or user-specified value), as well as volumes 2 forward and 1 back from these volumes. Furthermore, the voxel-specific mean framewise displacement (FD) and volume-level mean FD are output for accounting for head motion in group-level analyses.

In addition to head motion correction, DPARSF V3.2 also provides regression of other nuisance variables, including white matter (WM) and cerebrospinal fluid (CSF) signal regression. WM and CSF masks could be generated in two ways: 1) setting a probability threshold (e.g., 0.99) on one's own tissue segmentation maps based on his/her structural image; 2) using SPM's a priori tissue probability maps (empirical thresholds: 90 % for WM mask and 70 % for CSF mask). Other than mean signal regression, the CompCor method (Behzadi et al. 2007), which was reported to be advantageous (Chai et al. 2012), is integrated. This extracts the first 5 principal components from a combined WM/CSF mask.

Additionally, DPARSF V3.2 can distribute the processing of multi-subject datasets to different CPU cores with the MATLAB parallel computing toolbox. It supports defining regions of interests interactively based on a participant's T1 image in native space. It resamples the mask files automatically if the dimensions mismatch those of the functional images. Before co-registration of structural images and functional images, skull strip can be performed on each modality to improve the accuracy of registration.

Standardization Module

To reduce the impact of many sources of nuisance variation impacting on R-fMRI measures (Yan et al. 2013b), we have provided several post-hoc standardization approaches in the standardization module.

- 1) Mean Regression: calculate the mean across all brain voxels for a given R-fMRI measure of each participant and then regress out the mean at the group-level.
- 2) Mean Regression + SD Division: calculate the mean and standard deviation across brain voxels for each participant, and then regress out the mean at the group-level. The residual value at each voxel is divided by the standard deviation.
- 3) Mean Regression + log SD Regression: a mean regression + log SD regression model was employed to accurately address both additive and multiplicative noise in a two-step regression model. Please see Yan et al. (2013b) for more details.
- 4) Z-standardization: calculate the mean and standard deviation across brain voxels for a given R-fMRI measure, then subtract the mean from the value at each voxel and divide the value at each voxel by the standard deviation.
- 5) Mean Division: calculate the mean across brain voxels for a given R-fMRI measure and divide the value at each voxel by the mean.
- 6) Mean Subtraction: calculate the mean across brain voxels for a given R-fMRI measure and then subtract the mean from the value at each voxel.
- 7) Median - inter-quartile range standardization: calculate the median and the range between 25 and 75 % percentile (interquartile range), then subtract the median from the value at each voxel and divide the value at each voxel by the interquartile range.
- 8) Rank standardization: each voxel is ranked according to its value within the brain, and the rank value is assigned to each voxel.
- 9) Quantile standardization: each voxel is ranked according to value for each participant, and then the average value across participants of each rank is computed. The voxel with the same rank for each participant is assigned that average value.
- 10) Gaussian function fit normalization. First fit a Gaussian function with parameters of mean, standard deviation, and area to the distribution of values (restricted to full width at half maximum), then subtract the estimated mean from the value at each voxel and divide the value at each voxel by the estimated standard deviation (Lowe et al. 1998).

Test-Retest Reliability

Before using R-fMRI measures to characterize inter-individual differences, TRT reliability should be evaluated (Shehzad et al. 2009; Zuo et al. 2010a; Zuo and Xing 2014). Intra-class correlation (ICC) models the ratio of intra-individual variability and inter-individual variability (Muller and Buttner 1994; Shrout and Fleiss 1979), and has been widely used in fMRI literature to measure the TRT reliability (see a review in (Zuo and Xing 2014)). Usually a group of subjects are scanned several times to test the TRT reliability, and ideally we would like to investigate the interchangeable measures scanned at different occasions in time. However, given the presence of novelty, adaptation, learning and practice effects of fMRI scanning, the data scanned is not interchangeable across time. We thus have provided 2 forms of voxel-wise ICC calculation in the TRT reliability module. The first form is ICC(1,1) (Shrout and Fleiss 1979) using an ANOVA model: each target is measured by a different set of raters, randomly selected from a population of raters. The second form is ICC(3,1) (Shrout and Fleiss 1979) using linear mixed models: each target is measured by the same raters and these raters are the only raters of interest. The first form gives stricter ICC values as it assumes the measures are interchangeable. The second form is more suitable for this non-interchangeable situation of fMRI scanning, while needs longer computation time as linear mixed models are used. In practice, the two forms result in highly similar spatial patterns or regional differences across most conditions, while the second form avoids negative ICC values presented in the first form.

- 1) Form 1: ICC(1,1) with ANOVA model:

$$ICC = \frac{MS_b - MS_w}{MS_b + (k-1)MS_w}$$

Where MS_b is the between-subject mean square and MS_w is the within-subject mean square for each index, k is the number of scanned times (Shrout and Fleiss 1979).

- 2) Form 2: ICC(3,1) with linear mixed models

As described in Zuo et al., (2011; 2013), the two-level linear mixed model was applied to the decomposition of Y_{ij} (the value of the j -th participant's i -th measurement occasion):

$$Y_{ij} = \mu_{0j} + e_{ij}, \mu_{0j} = \mu'_{00} + e'_{0j}$$

where μ'_{00} is a fixed parameter and e'_{0j} and e_{ij} are independent random effects normally distributed with mean 0 and variances σ_e^2 and $\sigma_e'^2$. The term e'_{0j} is the participant effect and e_{ij} is the measurement error. The ICC was defined as

$$ICC = \frac{\sigma_e^2}{\sigma_e^2 + \sigma_e^2}$$

The variance terms were estimated with the restricted maximum likelihood (ReML) approach.

Of Note, we recommend the TRT reliability module to be used combining with paired t-tests (through the Statistical Analysis module). As mentioned above, systematic differences (presence of novelty, adaptation, learning and practice effects etc.) may exist between the fMRI data scanned at different occasions of time, these can be examined by paired t-tests. In the case of existing significant paired differences, the results can be reproducible although the ICC values are low.

Quality Control

To facilitate quality control, DPABI includes the following QC steps (Fig. 2).

1) QC for raw functional and structural images.

During preprocessing, there is a step for reorienting functional images and T1 images interactively. This step can help improve the accuracy of coregistration, segmentation and normalization, especially when the image's initial orientation is substantially inconsistent with a standard template. At the same time, it's an important QC step. DPABI pops up the related images for each subject automatically, allowing users to rate the quality of the structural and functional images and comment. For example, images with large distortions, head motion, drop out, ghosting, or structural abnormality can be screened. The scores and comments are used for thresholding the quality of the raw functional and structural images.

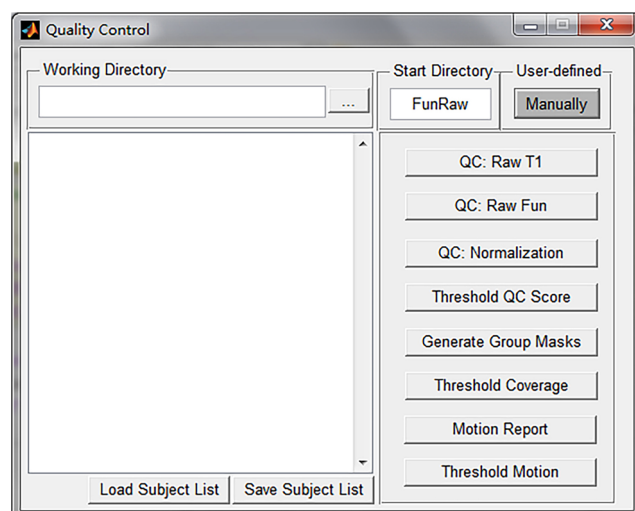


Fig. 2 The quality control user interface. Quality control steps include assessing the quality of raw functional and structural images, spatial normalization, coverage and head motion

2) QC for spatial normalization.

To make comparisons between subjects feasible, the individual brain is spatially coregistered (normalized) to standardized space. Ensuring adequate effects of normalization of each individual is important for drawing reliable conclusions. The DPABI QC module provides a convenient and efficient way to visually inspect coregistration. As demonstrated in Fig. 3, four yoked windows with normalized structural image, functional image, gray matter (GM) segmentation and a MNI template pop up. On top of each image, the GM/WM boundary is overlaid to better check the spatial normalization effect. The QC scores and comments can be stored through the rating window for later thresholding.

3) QC for spatial coverage and group mask generation.

Due to the limitation of how many slices can be included in functional scans and the variability in head size and shape (e.g., echo-planar imaging (EPI)), the spatial coverage of EPI raw images varies. The regions not covered in some subjects may produce spurious findings due to coverage differences. DPABI generates a coverage mask with a program equivalent to AFNI's 3dAutomask (https://afni.nimh.nih.gov/pub/dist/doc/program_help/3dAutomask.html). Specifically, the program estimates gradual clip values to set the background regions of the mean EPI volume to zero based on histogram gazing. Then the largest connected component of voxels above the clip values were selected to generate the coverage mask for a given subject, with erosion and filling the holes. Group masks could be generated including voxels, which are present in a minimum percentage (e.g., 95 %) of participants. Subjects who overlap with the group mask less than a certain amount (e.g., 2*SD under the group mean overlap) can be excluded.

4) QC for head motion.

Beyond the head motion correction strategies discussed in “Advanced DPARSF” section, excluding subjects with excessive head motion from statistical analysis is also essential. DPABI generates an Excel file including the following head motion metrics for each subject: max (or mean) absolute translations and rotations, mean RMS, mean FD Van Dijk (Van Dijk et al. 2012), mean FD Jenkinson (Jenkinson et al. 2002), mean FD Power (Power et al. 2012a), number of time points with mean FD Power > 0.5 (or 0.2), percentage of time points with mean FD Power > 0.5 (or 0.2). DPABI can pop up the head motion time course for each subject, and could threshold subjects according to their motion metrics. These quality control steps allow users to select those subjects with good quality and generate a subject list for statistical analysis.

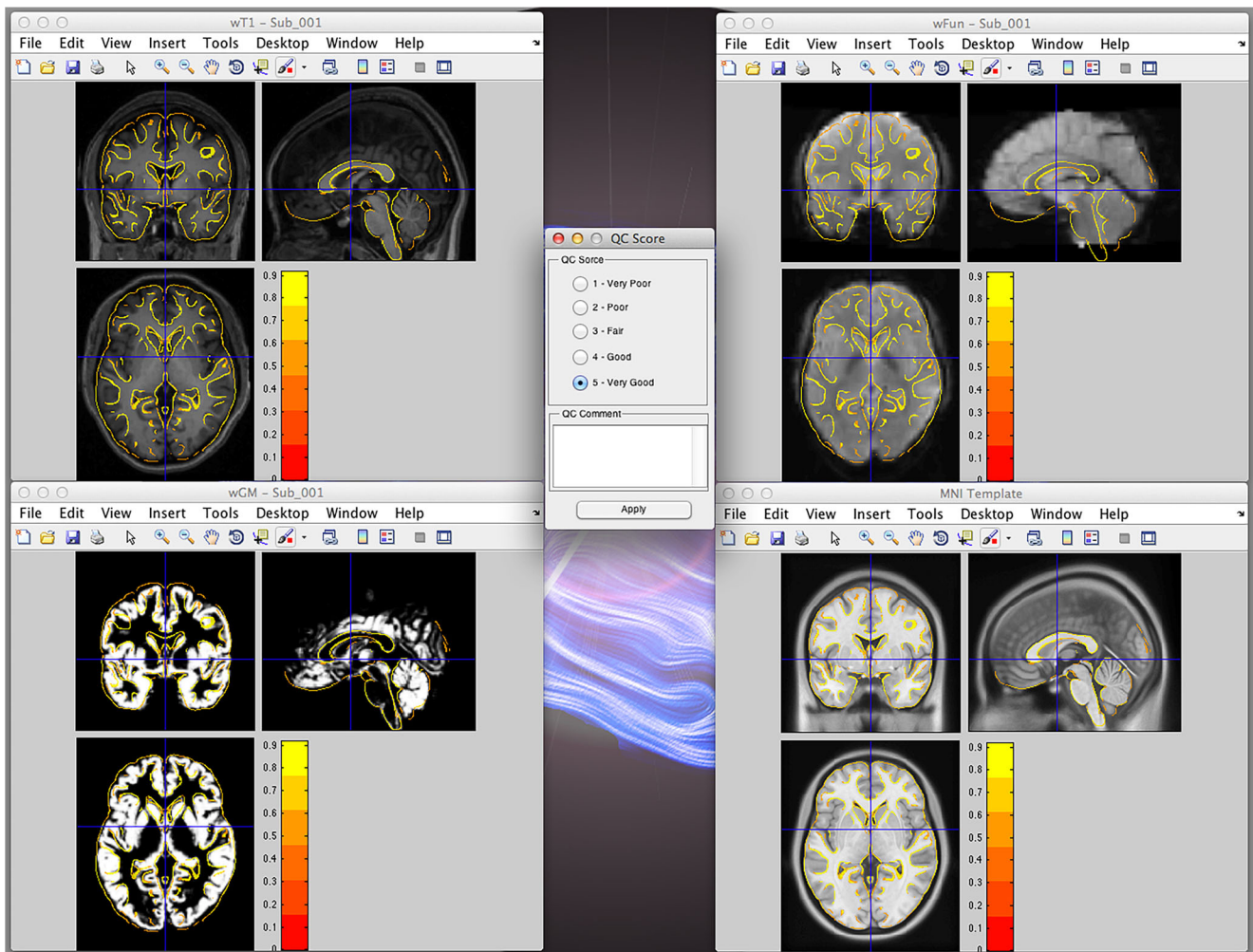


Fig. 3 Quality control for spatial normalization. *Upper left*: normalized structural image with the standard grey matter/white matter boundary overlaid; *upper right*: normalized functional image; *lower left*:

normalized gray matter segmentation; *lower right*: MNI template; *middle*: interface for users to give QC scores and comment

Animal Modules

To facilitate R-fMRI research on animal models, DPABI includes a module for macaque monkeys and a module for rats.

For monkey data processing, most of the steps are similar as for humans, except for spatial normalization. The individual monkey structural image is coregistered to the functional image. Then, using the monkey prior tissue probability maps generated by McLaren et al. (2009), the transformed structural image is segmented into GM, WM and CSF through unified segmentation (Ashburner and Friston 2005). The functional volumes are normalized to the 112RM-SL space using the normalization parameters estimated during unified segmentation (McLaren et al. 2009, 2010). The remaining steps follow the human processing pipeline.

For rat data processing, the voxel size of the rat structural and functional images is usually too small for the standard neuroimaging software packages, which were designed for human imaging. Thus, DPABI offers a function (Voxel Size

Augmentor) to augment (usually 10 times) the voxel size. A rat T2 template (Schwarz et al. 2006) is used for spatial normalization.

Other Utilities

DPABI provides a preprocessing pipeline for task-based fMRI, including DICOM to NIfTI, slice timing, realignment, nuisance regression, spatial normalization, and smoothing. The novel part of this module is that it provides a better batch pipeline for task-based fMRI data preprocessing, thus requires less manual operations as compared with SPM's native batch system. This module only provides preprocessing function; users could perform further analyses (e.g., GLM analysis and psychophysical interaction analysis) using other software (e.g., SPM) based on these preprocessed data. In addition, it also supports VBM analysis including segment structural images into GM, WM and CSF (Ashburner and Friston 2005) and then uses the Diffeomorphic Anatomical Registration

Through Exponentiated Lie algebra (DARTEL) tool (Ashburner 2007) to compute transformations from individual native space to MNI space. The only novel part of this module is the batch processing system which required less manual operations than SPM's native VBM processing and VBM8 (by Christian Gaser).

DPABI also offers a statistical analysis module supporting common statistical models, which are also available in FSL, AFNI, SPM and REST etc. The novel part of this module in DPABI is the post-hoc comparison correction over group pairs after ANOVA analysis. Most neuroimaging studies perform pairwise two-sample t tests after ANOVA analysis, which provides no protection against multiple comparisons over group pairs. In DPABI, we introduced post-hoc procedures including Tukey-Kramer (honestly significant difference), Bonferroni, Dunn-Sidak and Scheffe control procedures. The corrected p values under a given control procedure for comparing group means of any pairs were calculated (e.g., through Studentized Range statistic for Tukey-Kramer correction) with the same route as MATLAB command *multcompare* (<http://www.mathworks.com/help/stats/multcompare.html>). The p maps were then converted to Z maps according to the Normal inverse cumulative distribution function (*norminv*), with the sign of group mean differences applied. Using the Z maps, users can perform Gaussian random field (GRF) theory or AlphaSim correction to correct multiple comparisons over voxels further. To the best of our knowledge, this is the first neuroimaging package that offers post-hoc control procedures over multiple group pairs. Furthermore, during statistical analysis, DPABI estimates smoothness based on 4D residuals and writes to the NIfTI headers, similar to FSL's *smoothest* program. The smoothness is used for GRF correction for multiple comparisons or Monte Carlo simulation (AlphaSim). Many studies used smooth kernel in preprocessing to estimate the biggest cluster size with GRF or AlphaSim correction. However, the effective smoothness is bigger than the applied because the pre-smooth image has implicit smoothness. Simply inputting Gaussian smoothing kernel that was applied during preprocessing to AlphaSim is incorrect (Bennett et al. 2009). DPABI prevents that kind of errors by performing AlphaSim or GRF correction based on the estimated effective smoothness.

Finally, the results can be viewed in the DPABI Viewer (revised from *spm's* function *spm_orthviews*). DPABI Viewer integrates key functionalities in FSL's *fslview*, AFNI, MRICroN, REST slice viewer, XjView and BrainNet Viewer, and provides interfaces to them. DPABI Viewer allows displaying multiple images in different layers, as well as creating region of interest masks based on atlases or statistical maps. DPABI viewer reports the structure names of clusters and generates reports based on XjView (by Xu Cui, <http://www.alivelearn.net/xjview8/>). The viewer also provides an interface to the surface displaying tool, BrainNet Viewer

(Xia et al. 2013), thus producing a volume map and a corresponding surface map with the same color and range settings.

In addition, DPABI provides several utilities. DPABI includes a novel Image Averager: when a subject has several repetitions of structural images, this module will coregister them together and make an average image, which has higher signal-to-noise ratio. A T1 Image Defacer is included, with similar function to MRI Defacer (https://www.nitrc.org/projects/mri_deface/). Here we implemented a batch defacing system for a future data-sharing project, which needs removing the privacy of the structural images. DPABI has also adapted the utilities of DICOM Sorter, Image Calculator and Image Reslicer, which has been implemented and described in REST (Song et al. 2011).

Illustration

We have provided detailed online video course to describe the usage of DPABI and how to choose processing steps, which is available at <http://rfmri.org/Course>. We have also created a wiki website for DPABI (<http://wiki.rfmri.org>), that all the users (registered at the R-fMRI Network) can edit any item of the wiki, gradually forming a detailed knowledge base for DPABI and R-fMRI by the community effort. To better demonstrate the functions of the software, here we illustrated the data processing steps with a practical R-fMRI data.

Data

We used the publicly available imaging data from the 1000 Functional Connectomes Project (RRID:SCR_005361, all data are available at http://fcon_1000.projects.nitrc.org) to demonstrate the functions of DPABI. The primary results were reported in Yan et al. (2013b); here we illustrate DPABI usage with re-generated figures. Datasets of 944 subjects from 18 sites, including an R-fMRI run and a T1-weighted anatomical image, were used for analyses.

Preprocessing

All preprocessing was performed using the advanced DPARSF module V3.2. All volume slices were corrected for different signal acquisition times. Then, the time series of images for each subject were realigned. Individual structural images (T1-weighted MPRAGE) were co-registered to the mean functional image after realignment. The transformed structural images were then segmented into GM, WM and CSF (Ashburner and Friston 2005). To remove the nuisance signals, the Friston 24-parameter model (Friston et al. 1996) was utilized to regress out head motion effects from the realigned data. The signals from WM and CSF were regressed out to reduce respiratory and cardiac effects. To better control

head motion effects, each bad time point with high head motion was included as a regressor. The bad time points were defined as volumes with FD (Jenkinson) > 0.2 mm as well as volumes 2 forward and 1 back from these volumes. The Jenkinson definition of FD (Jenkinson et al. 2002) was used due to its consideration of voxel-wise differences in motion in its derivation (Yan et al. 2013a). In addition, linear and quadratic trends were also included as regressors since the BOLD signal exhibits low-frequency drifts. The DARTEL tool (Ashburner 2007) was used to transform the functional data from individual native space to MNI space. Spatial smoothing (FWHM kernel: 4.5 mm) were applied to the functional images (except for ReHo and degree centrality calculation). Further, temporal filtering (0.01–0.1 Hz) was then performed on the time series except for the frequency-based R-fMRI indices: ALFF and fALFF.

The following R-fMRI-based indices of intrinsic brain function were examined: 1) ALFF (Zang et al. 2007) / fALFF (Zou et al. 2008); 2) ReHo (Zang et al. 2004); 3) VMHC (Anderson et al. 2011; Zuo et al. 2010b); 4) seed-based correlation analysis of the posterior cingulate cortex (PCC: 0, –53, 26; 10 mm diameter sphere) (Satterthwaite et al. 2012; Van Dijk et al. 2012; Yan et al. 2013a); and 5) network degree centrality (Buckner et al. 2009; Zuo et al. 2012). A comprehensive review of these rfMRI metrics is provided elsewhere (Zuo and Xing 2014). Of note, for VMHC calculation, an extra preprocessing step was performed to improve the correspondence between symmetric voxels. The individual structural image was reregistered to a group averaged symmetric template, and then this refined transformation was applied to the functional data before VMHC calculation (Zuo et al. 2010b).

Quality Control

The quality control module was used to score datasets and exclude those that did not meet quality criteria. By rating the raw functional and structural images, as well as their normalization effects, using the rating function in DPABI, 49 subjects were excluded, leaving 895 subjects to be analyzed. A group mask was generated by including voxels present in at least 90 % of participants (Fig. 4). Thirty-four subjects with overlap with the group mask less than $2 \times \text{SD}$ under the group mean overlap (threshold: 92.2 %) were excluded, resulting in 861 subjects. Finally, using the head motion thresholding function, 53 subjects with motion (Mean FD Jenkinson (Jenkinson et al. 2012)) greater than $2 \times \text{SD}$ above the group mean motion (threshold: 0.192 mm) or with > 50 % of time points being scrubbed were excluded, resulting in 808 subjects. Those 808 subjects passing quality control were forwarded to further analysis.

Standardization

For the R-fMRI measures, acquisition-, experimental-, environment- and participant-related variations can introduce unintended variations in the measurements (Friedman and Glover 2006a; Shannon et al. 2013; Vanhoutte et al. 2006; Yan et al. 2009, 2013b). Measurement standardization is needed to reduce the impact of many sources of nuisance variation on the R-fMRI measures. In our previous study (Yan et al. 2013b), we have compared an array of standardization approaches and found mean-regression based group-level post-hoc standardizations were advantageous. Here, these two standardization approaches were taken as examples via the standardization module. 1) Mean Regression: calculate the global mean for each subject and then regress out the mean at the group-level. 2) Mean Regression + SD Division: The residual value after mean regression is divided by the standard deviation for each subject.

Test-Retest Reliability

The TRT reliability module was used to evaluate the reliability of R-fMRI measures after standardization. The NYU TRT dataset, which contains three scans for 25 subject, was used to evaluate the ICC; the first scan (scan 1) was collected 5–16 months (mean $\pm$ SD = 11 ± 4 months) before two subsequent scans (scans 2 and 3), which were collected in a single session ~45 min apart. We evaluated inter-session reliability as the ICC between scan 1 and the average of scans 2 and 3. ICC values were derived by linear mixed models (LMMs) as per Zuo et al. (2013). As demonstrated in Fig. 5 (surface maps were produced by BrainNet Viewer (Xia et al. 2013), through an interface in the DPABI Viewer), ALFF without standardization showed very low TRT reliability, which would prevent its usage. With standardization, whether with mean regression or mean regression & SD division, ALFF showed moderate TRT reliability. All the other measures showed moderate TRT reliability, slightly reduced by standardization. The reductions in TRT reliability could be resulted from the significant TRT reliability of the standardization parameters (global mean and global SD) (Yan et al. 2013b). Of note, nuisance signals (e.g., head motion) can be pretty reliable and could artifactually increase the reliability of the metrics, thus these reductions in reliability should not be viewed as discouraging (Yan et al. 2013b; Zuo et al. 2010b).

Statistical Analysis

The Statistical Analysis module was used to examine inter-individual differences in R-fMRI measures related to age while taking the confounding effects of sex, site and mean FD into account. Age effects were estimated by the t-value of the corresponding regressor in the general linear model.

Fig. 4 The group mask generated by including voxels present in at least 90 % of participants

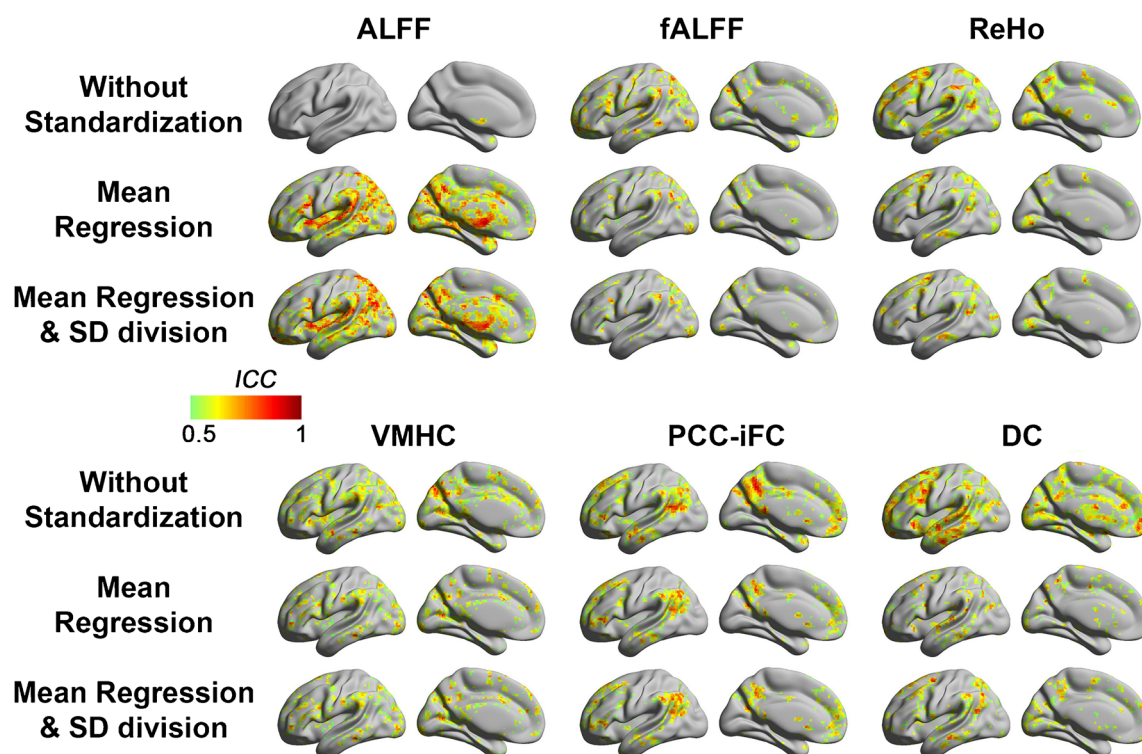
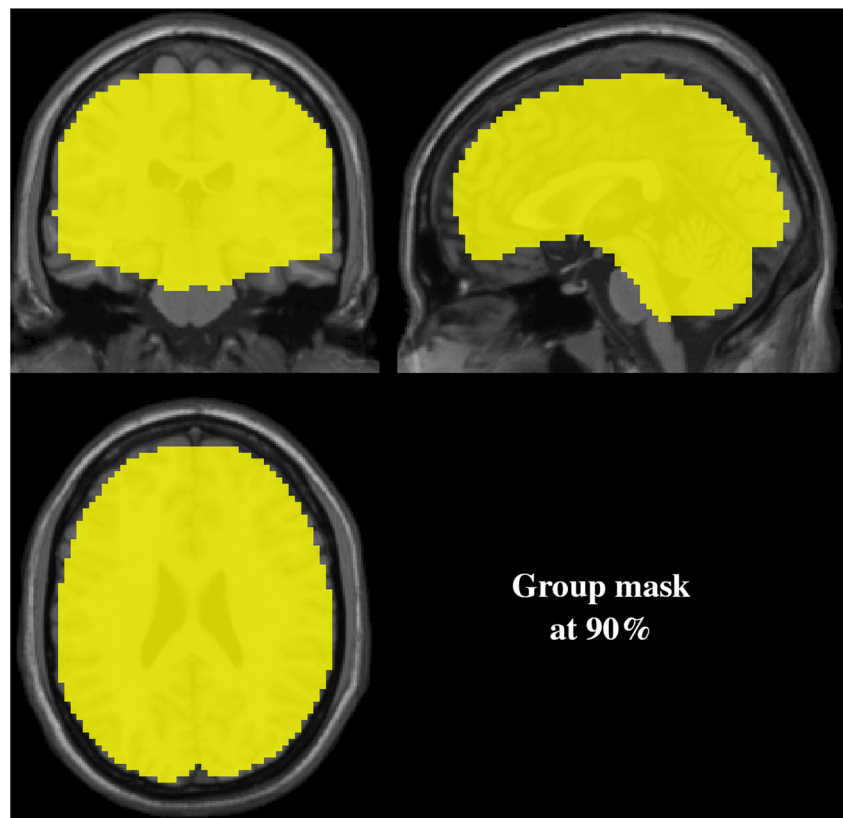


Fig. 5 The test-retest reliability (indexed by intra-class correlations) of the R-fMRI measures with and without standardization

GRF correction for multiple comparisons was applied for the voxel-wise maps of R-fMRI derivatives. In appreciating recent reports on enhancing the primary threshold (p value at individual voxel level) to $P < 0.001$ (Eklund et al. 2015; Woo et al. 2014), we have set voxel-level $Z > 3.1$, cluster-level $P < 0.05$ for multiple comparison correction. Stringent primary threshold ($P < 0.001$) increases spatial specificity and better prevents false positives, while suffers more from false negatives and reduces the probability of overlapping for the significant clusters across studies. As demonstrated in Fig. 6, there were negative age associations with unstandardized R-fMRI measures across brain regions for most R-fMRI measures. After standardization, the distributed negative-relationships were removed; regions with remaining positive age effects are shown. For the standardization of mean regression & SD division, each measure has at least one significant cluster, while the cluster size is small given the stringent primary threshold.

Discussion

Here we described the newly developed toolbox, DPABI, which includes an advanced edition of DPARSF (V3.2), modules for head motion control, standardization, TRT reliability, quality control and rat/monkey data analysis pipelines. Processing of other imaging modalities, including

preprocessing modules for task-based fMRI and voxel-based morphometry analysis are included. DPABI also provides modules for statistical analysis, an MRI viewer and a series of image utilities.

We illustrated DPABI usage by applying it to a standardization project based on public datasets. With the integrated head motion control functions, DPABI makes it straightforward to evaluate results using strict motion correction strategies, to minimize spurious findings from motion effects. The standardization module minimizes the influence of nuisance variables on R-fMRI results. With the TRT reliability module, investigators can first test the reliability of their measures before investigating inter-individual differences. The quality control module helps users closely inspect their data, as adequate quality control is essential for establishing reliable and valid results. The statistical analysis module allows users to perform statistical analysis and multiple comparison corrections conveniently. Results can be viewed and high quality images exported easily using the viewer. Furthermore, the animal processing modules allow researchers to conveniently examine pharmacological/behavior manipulation effects and explore the underlying physiological mechanism and pathology related changes in R-fMRI data through animal studies.

DPABI is designed so that data analysis will require fewer manual operations, be less time-consuming, have a lower skill

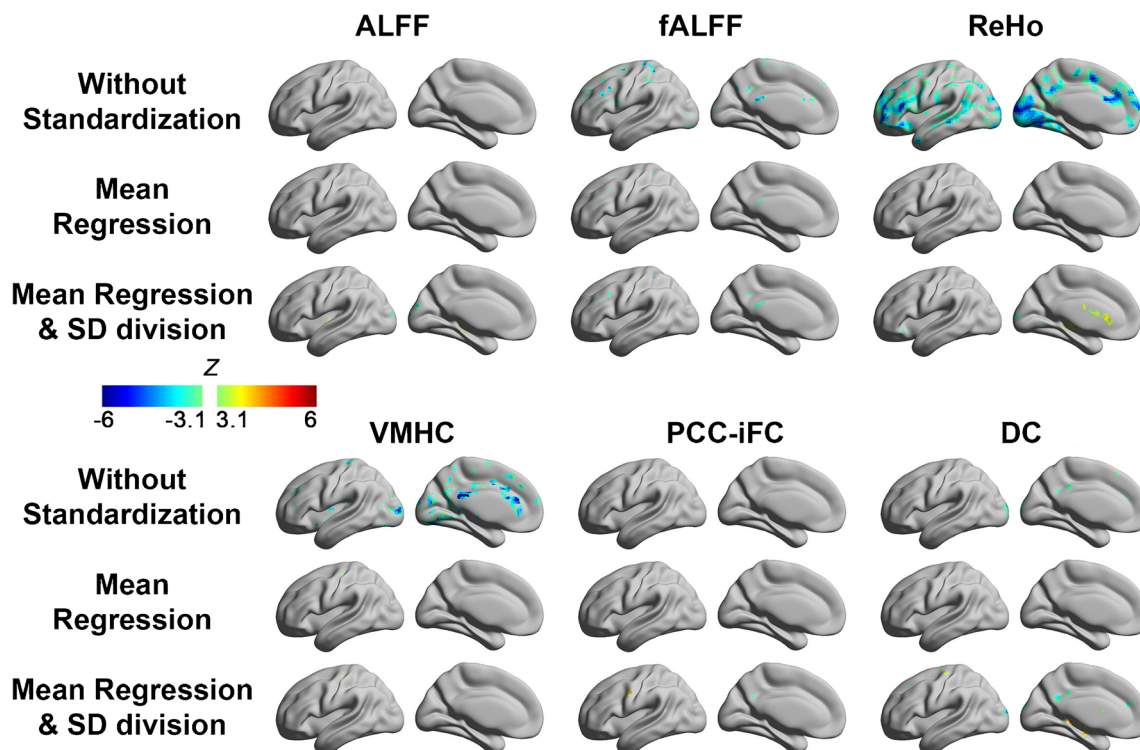


Fig. 6 The age effects of R-fMRI measures with and without standardization. *Warm colors* indicate positive correlations with age and *cool colors* indicate negative correlations with age

requirement, smaller possibility of inadvertent mistakes, and greater comparability across studies. Given these potential advantages, DPABI could set a standard data processing stream for R-fMRI data. If that were to occur, having data processed with the same procedure through the same software, should yield more comparable results across sites and an increase in their reproducibility (Poldrack and Poline 2015). The next step for DPABI is to create a data-sharing platform, and encourage users to share R-fMRI maps processed with DPABI. A large database of R-fMRI maps of subjects across various brain disorders could be accumulated through such a platform. Making R-fMRI maps available reduces computational demands and neuroimaging preprocessing skill requirements, thus allowing researchers from multiple disciplines to utilize big data to perform innovative studies. Moreover, such accumulated big data furthers the open science agenda to propose and address critical scientific questions (Milham 2012).

Of note, DPABI runs on MATLAB, a commercial software with a costly license, which could be prohibitive for some users. Although there is a parallel computation toolbox available with MATLAB, it doesn't suit the recently emerging high performance computing clusters well. Given those limitations that might prevent a wider usage of DPABI, we plan to make a stand-alone version of DPABI and later adapt it to Python in future.

In sum, we assert that DPABI is user-friendly, allowing users to process data with a graphic user interface (GUI) (command line alternative available), perform statistical analyses and view the results. We hope this open-source toolbox will assist novices (via the user-friendly GUI) and expert users (by efficient command line) and continue to support advancing R-fMRI methodology and its application to clinical translational studies.

Information Sharing Statement

The source code of DPABI (RRID:SCR_010501) is openly shared at <http://rfmri.org/DPABI>.

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Compliance with Ethical Standards

Conflict of Interest DPABI is hosted at the R-fMRI Network (<http://rfmri.org>), which is a non-commercial resource for the R-fMRI community.

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